

Electrical Design Review

Type	Design Review	**Document no.**	VTX-EE-DR-001
Version	1.0	**Status**	For Review
Owner	Electrical Team	**Approver**	CTO Hardware
Author	Electrical Team	**Date**	November 3, 2026

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BMS

1 Design overview

The BMS is in charge of monitoring and protecting the 4S LiPo battery in the AUV. It measures battery current and cell voltages, controls the battery output and reports faults to the rest of the submersible. [SPEC, Sec. 17.1, 17.3–17.4, 17.8]

Development started with the requirements and the previous BMS design. The circuit was then divided into functional blocks: battery monitoring, MCU, communication, power supplies and clocks. These were developed in tandem with the MCU pin allocation and continuously revised as the schematic progressed.

At the time of writing, I have just completed the schematic step, and am in the process of the PCB Layout and wire-routing. The main challenge so far has consisted of making the blocks work together, particularly their supply voltages, startup behaviour and our focus on low-power operation.

2 Design inputs and constraints

The design follows *Team 2027 Electrical System Technical Specification Rev 23*. This specifies the shared MCU, electronics protection, isolated CAN interface and development connections. [SPEC, Sec. 4.3, 4.7–4.18]

The BMS must also reuse the previous BMS-IC family and minimise power consumption, since it remains connected to the battery outside the submersible. [SPEC, Sec. 17.5, 17.9]

3 System architecture

The BMS-IC handles battery measurements and protection. The MCU reads this information over SPI and communicates with the rest of the AUV over CAN. USB and debug interfaces provide access during development.

The electronics supply is protected by an eFuse and regulated to 3.3 V and 5 V. USB is intended as an alternative supply for the various BMS components.

It should also be noted that this year's BMS does not carry the capacity to charge the battery pack directly. Therefore, all charging of the 4S battery must be conducted with an external LiPo charger.

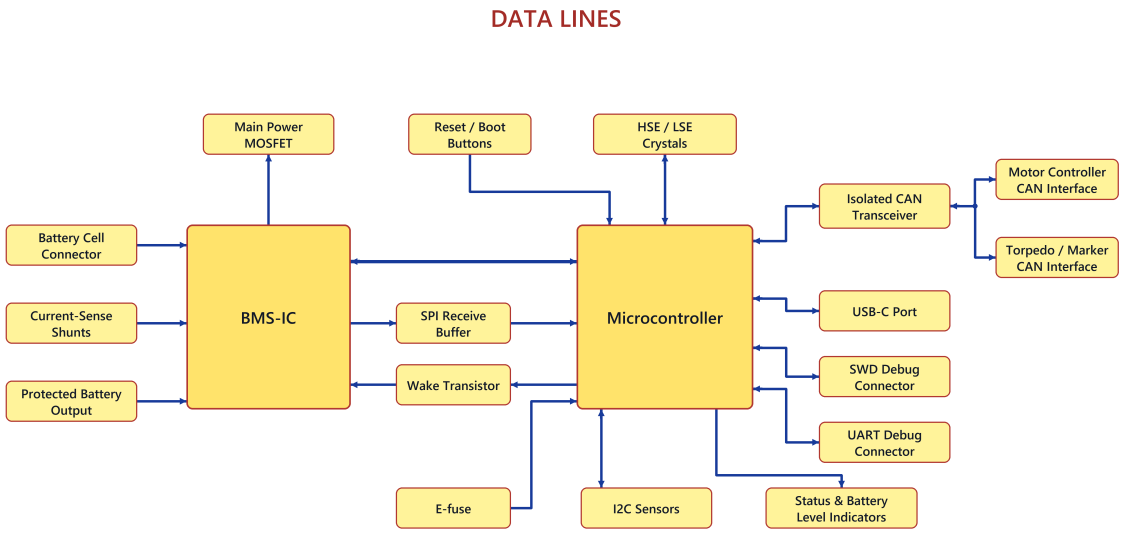


Figure 1: Illustrative flowchart of BMS data-lines.

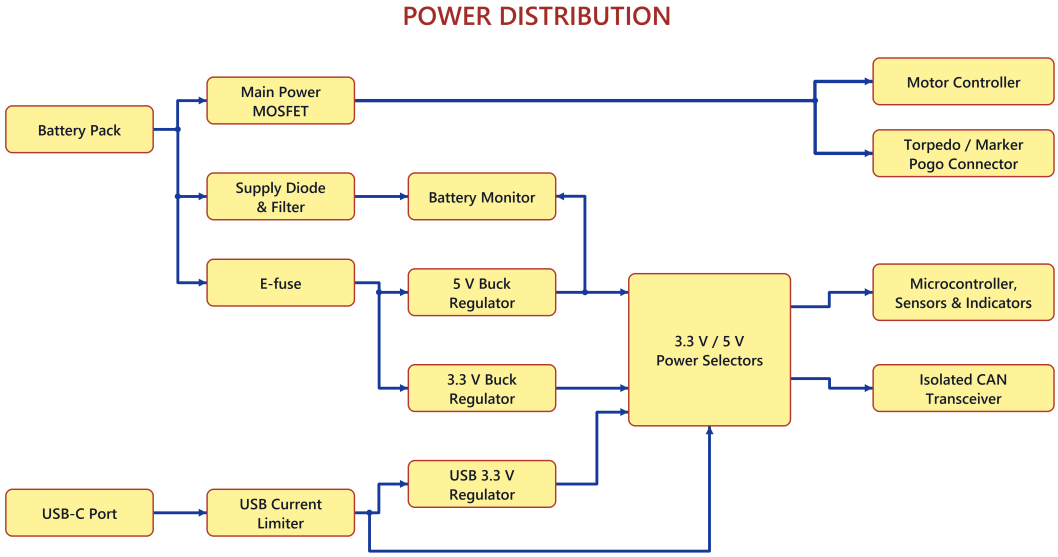


Figure 2: Illustrative flowchart of BMS Power-lines.

4 Detailed design

4.1 Microcontroller

The STM32C542CCT6 handles configuration, communication and indicators, following the shared MCU requirement. The main challenge is coordinating the pin allocation and firmware with SPI, USB, debugging and FDCAN2 on PB5/PB6. It is also critical that the startup states and sleep behaviour support our low-power requirement. [SPEC, Sec. 4.3, 4.11, 4.17, 17.9] [BMS-1]

4.2 BMS-IC

The BQ7694201PFBR measures cell voltages and shunt current and controls the main battery switch. The important remaining work is to configure the four-cell measurement arrangement, protection limits, SPI/CRC communication and wake-up behaviour. Battery-mounted temperature sensing also remains unresolved. [SPEC, Sec. 17.1, 17.3–17.6] [BMS-2]

4.3 CAN communication

The ADM3057EBRWZ provides the required isolated CAN interface. It uses a 5 V supply with 3.3 V MCU logic. This introduced a separate 5 V supply requirement and made the isolated supply and ground connections important parts of the design. Standby operation remains to be resolved due to our current inability to monitor the board's power budget. (26th October 2026) [SPEC, Sec. 4.10–4.11, 17.9] [BMS-3]

4.4 USB-C power and logic

The USB block uses a TPS2553 current limiter, an ADP3338 3.3 V regulator, two TPS2113A source selectors and an SN74LVC1G17 detection buffer.

The challenge is allowing battery and USB operation without backfeeding either source. Source switching, current consumption and regulator heating remain to be tested. The active source-selection solution also needs acceptance against the diode arrangement in the specification and remains an unknown until further measurements can be conducted. [SPEC, Sec. 4.12–4.13] [BMS-4]

4.5 Electronics eFuse

The TPS26400PWPR protects the electronics supply before voltage regulation. It does not carry the main motor current. The shared undervoltage and overvoltage network follows the specification, while current limiting and startup timing must be checked for this board. [SPEC, Sec. 4.7–4.9] [BMS-5]

4.6 HSE and LSE clocks

The last components I wanted to highlight so far in the development of the BMS are the clocks. The design uses a 16 MHz Würth 830108207109 crystal and a 32.768 kHz Würth 830054236 crystal. These are passive crystals used with the MCU oscillator circuits. Our choice of having two distinct oscillators is a direct result of our goal to reduce power consumption as much as possible. By utilising a slower clock for some of our operations, we hope to reduce our load on the battery while plugged in. [BMS-6] [BMS-7]

Motorcontroller

1 Design overview

The motorcontroller PCB is the interface between the ORIN pc and the ESC The ESC takes in the D-Shot protocol, which we will imitate with PWM additionally the computation of PWM values will be done on the microcontroller instead of ORIN to offload computing. additionally the motorcontroller will further ensure that the current and voltage that reaches the ESC is safe and reliable. Additionally the motorcontroller PCB enacts the killswitch for the thrusters in a hardwired fashion. A diagram of the overall design can be found in figure 3.

2 Design inputs and constraints

2.1 Key requirements

All shared requirements under section 4 in the Electrical System Specifications [SPEC] apply to the motorcontroller PCB. All requirements under section 18 apply to the motorcontroller PCB. The requirements include, but are not limited to:

Table 1: Design-driving requirements

Source ID	Requirement / target	Design impact
REQ-7.1	Circular PCB with max 75 mm diameter	Strict component placement; prioritized core features over optional additions.
REQ-18.3	Monitor current usage (motors & components)	Requires integration of current-sensing circuitry or smart power switches.
REQ-18.4	Monitor battery voltage for calculations	Requires voltage divider and ADC input channels on the microcontroller.
REQ-18.8	Hardwired power cut to thrusters	Integration of a physical hardware-level safety shutdown circuit.
REQ-8	CAN-bus communication with the system	Requires an onboard CAN transceiver and appropriate termination/routing.

2.2 Rationale

The 75 mm diameter size constraint is dictated by the physical mechanical enclosure limits of the underwater vehicle's housing, ensuring the motorcontroller PCB mounts securely without interfering with surrounding hardware. Current and voltage monitoring are essential for telemetry, protection, and closed-loop control algorithms. The hardwired power cut is a critical safety requirement mandated by the RoboSub competition. Finally, CAN-bus communication is chosen for its robust, deterministic messaging across the distributed electrical system.

3 System architecture

The Opto-coupler is implemented according to the logic in 10.3 [SPEC] as to mitigate the influence of noise to the killswitch signal logic. Several Debugging interfaces: USB-C, UART,

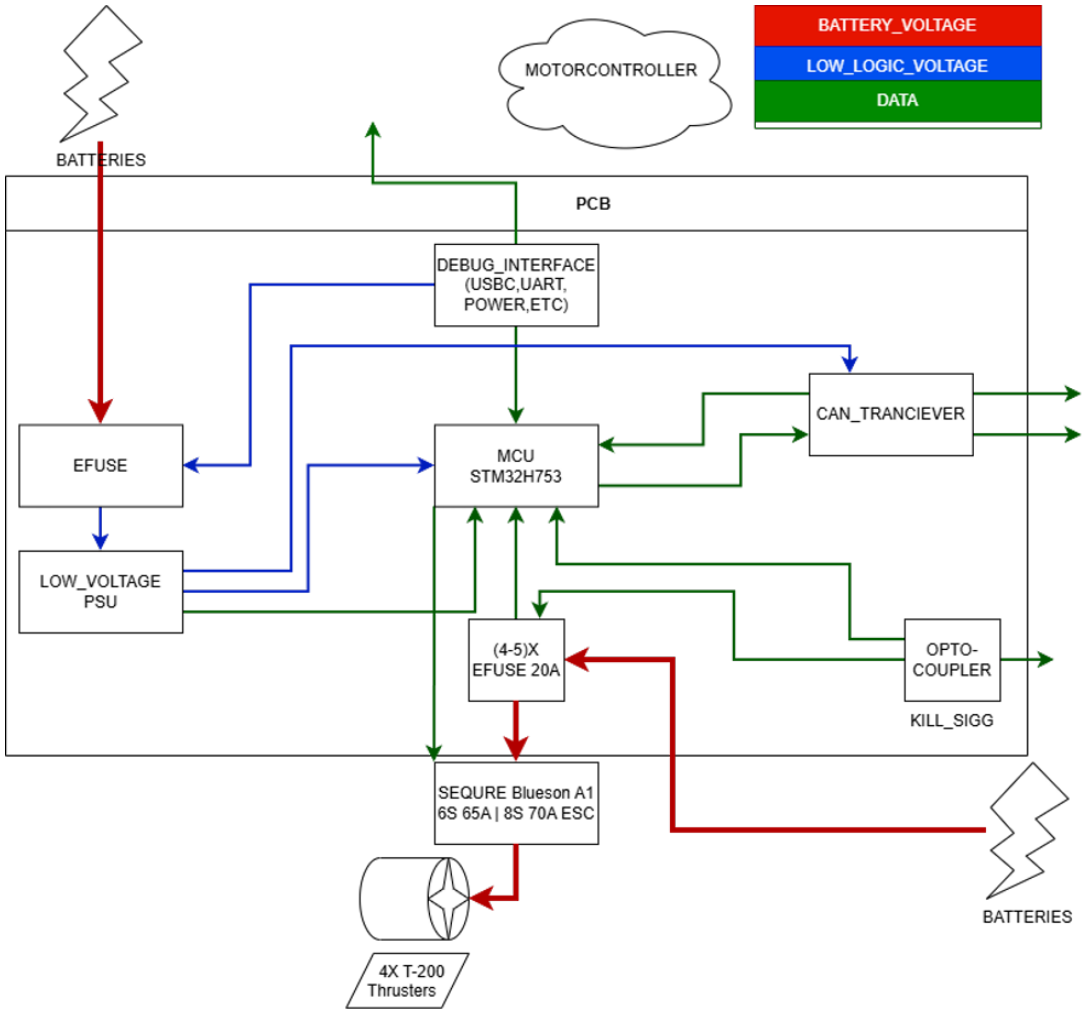


Figure 3: Motorcontroller PCB concept overview.

SWD to ensure maximum comfort for the embedder. We will also use a STM32H7 instead of the STM32C5 for we expect to need more computing power for the calculation algorithms.

4 Detailed design

4.1 General design

The STM32H753VIT6 was selected for its increased computing power. We need more computing power to imitate the D-shot protocol and calculate the desired PWM values from our read Battery-voltage levels. The thrusters require "higher" PWM values for lower battery voltage. The TCAN3414DR,173950336,TPS26400PWPR,830207659609 and BME280 were chosen as a part of the shared requirements, so that in the case of an accident we have multiples ready to go.

4.2 Calculations and margins

Calculations for the TPS26400PWPR [MC-2] are the same for all PCBs according to the requirements in section 4.8 of [SPEC]. The current-limit of the TPS26400PWPR is set to roughly 500**mA** this is well within what the small components will use and the same as what the 173950336 [MC-3] can deliver at maximum. This was found using the Calculator given by Texas instruments for this component [MC-4]. The rest of the components were given the component values specified in their respective datasheets.

4.3 Interface

The other boards on the CAN-bus interface with the motorcontroller through CAN and the microcontroller uses UART, I2C and D-shot to talk to its peripherals. It gets power from the BMS and delivers it to the ESC.

4.4 eFuses

As the PCB must channel peak 100A, 60-70A continuous we need eFuses that can handle these loads while also providing current monitoring and latch off capabilities under an overcurrent condition. For this a parallel coupling of two NCP81296MNTXG [MC-1] were chosen. These can together continously carry 100A of current, well within the requirement. We also only need two of them reducing space costs.

4.5 Calculations and margins

The cutoff current was set to 52A while the eFuses pull down their DOC pin on 45A, resulting in $R_{CLREF} = \frac{V_{CLREF}}{I_{CL}} \approx 100\mathbf{k}\Omega$, where $V_{CLREF} = I_{CS}R_{CS} = 455\mu A \times 2000\Omega$ and $I_{CS} = 45A \times 10\mu A/A + 5\mu A$. The calculations are further described in the datasheet for NCP81296MNTXG.

4.6 Interface

The microcontroller has the ability to cut the power to the thrusters aswell, these commands are issued from the ORIN-computer through the CAN-bus. The power to the thrusters are taken directly from the BMS to ensure that the high current lines are as short as possible.

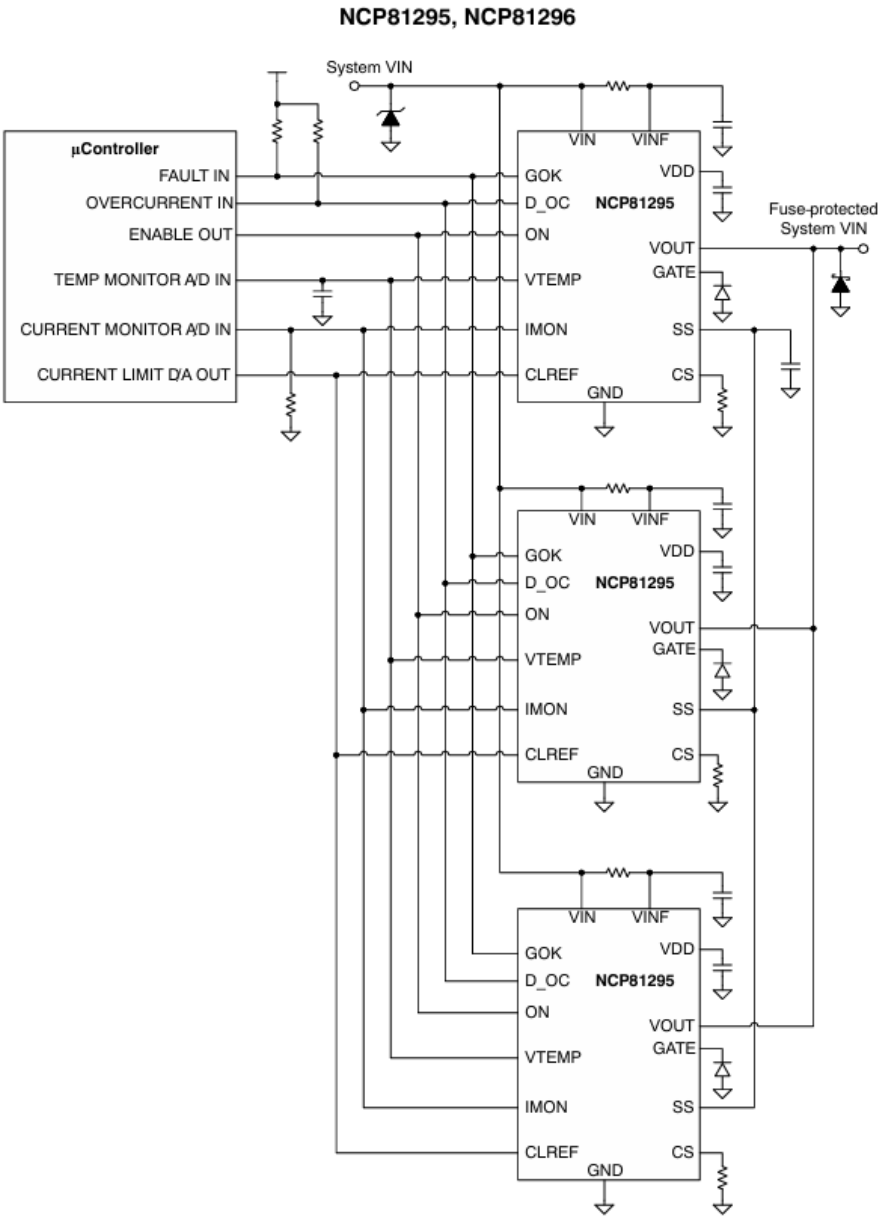


Figure 4. Application Schematic – Parallel Fuse Operation with Controller

Orin BOB

1 Design overview

The Orin BOB (breakout board) mounts, on the 40-pin expansion header of the Jetson AGX Orin and brings out I2C, SPI, UART, CAN, one PWM output and two GPIO (one in, one out) to connectors. All signals are galvanically isolated, so a wiring fault or ground loop on an external sensor cannot reach the Orin. The board is powered from the Orin header. The Orin, cables and external devices are outside the scope.

2 Design inputs and constraints

2.1 Key requirements

All shared requirements under section 4 in the Electrical System Specifications apply to the Orin BOB [SPEC, Sec. 4]. The most relevant are listed in Table

Table 2: Design-driving requirements (replace the placeholder row)

Source ID	Requirement / target	Design impact
REQ-001	Breakout I2C, SPI, UART, CAN, PWM and two GPIO	One isolator and one connector per bus
REQ-002	All external signals isolated from the Orin	Digital isolators with isolated supplies
REQ-003	Isolated CAN [SPEC, Sec. 4.10-4.11]	ADM30057E, same transceiver as the other PCBs
REQ-004	Powered from the Orin header, 1A max per rail	Low power budget on 3.3V and 5V

2.2 Rationale

The Orin is the most expensive part of the system and cannot be repaired at competition, so everything we plug into it goes through an isolation barrier. CAN must be isolated on every node according to the specification, and using the same transceiver as the other PCBs means wee share spare parts. The BOB has no power input of its own, so it has to stay well below the 1A rating of the Orin header pins.

3 System architecture

Each bus has its own isolator 4. The Orin side is supplied from the Orin 3.3V and 5V pins. U5 and U3 have built-in isolated DC/DC converters that make the 3.3V supply for the connectors. U5 also supplies the isolated side of the I2C. U6 has its own isolated supply and carries the PWM signal on its auxiliary channel. All connectors share one isolated ground (GND_ISO).

4 Detailed design

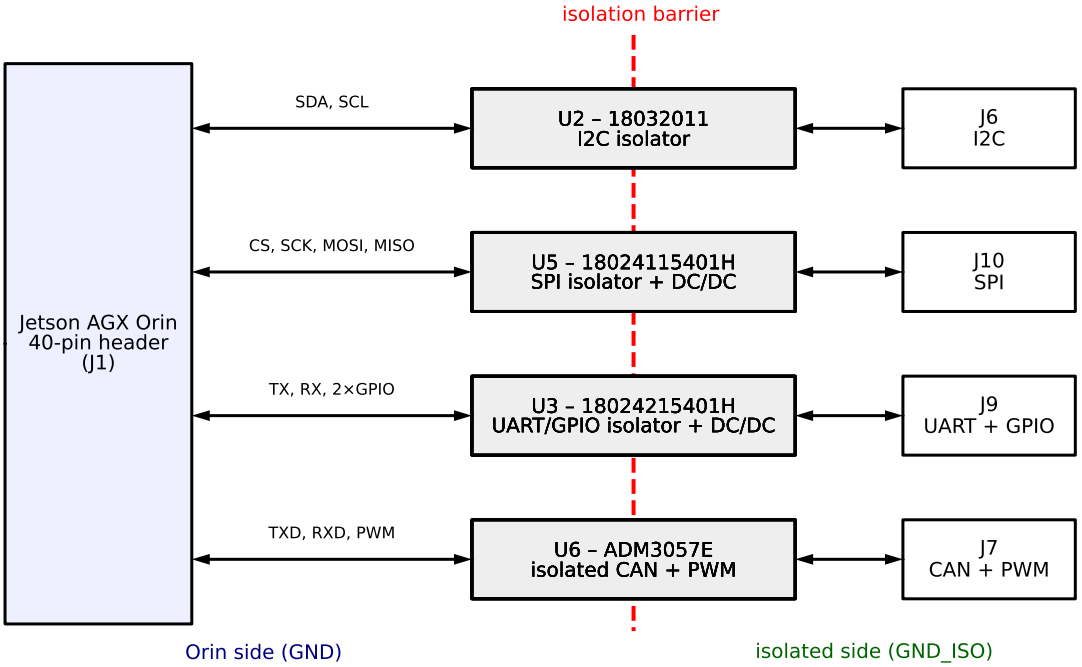


Figure 4: Orin BOB architecture.

4.1 Isolators

U5 and U3 are Wurth WPME-CDIP isoltors with an integrated 0.65 W isolated DC/DC. Y5 is the 3-forward/1-reverse version which fits SPI (CS, SCK, MOSI out, MISO in). U3 is the 2/2 version for UART TX/RX and the two GPIO. The H version defaults outputs high, so CS and UART stay idle if the Orin side is unpowered. SEL is tied to GND2 for a 3.3V output. Decoupling is $10\mu F$ on each side as in the datasheet.

U2 is a Wurth 18032011 I2C isolator with both SDA and SCL bidirectional, so slaves can stretch the clock. It has $4.7k\Omega$ pull-ups on both sides. U6 is the ADM3057E isolted CAN FD transceiver. VCC is 5V from the Orin and VIO is 3.3V. RS is tied to GND2 for full speed, and STBY and SILENT are left open (internal pull-downs, normal mode).

VISOOUT/GNDISO go through ferrite beads as recommended. R6 (120Ω) terminates the bus, so the bus, so the BOB must sit at the end of the CAN bus. The auxiliary channel (max 20kHz) carries PWM01 to J7.

4.2 Calculations and margins

The CDIPs converter deliver 75 mA at 3.3V in/ 3.3V out and 50-55mA at full data rate. The I2C pull-ups use

$$\frac{2 \cdot 3.3V}{4.7k\Omega \approx 1.4mA} \tag{1}$$

, leaving about 45mA for J6 and J10 together and 55mA for J9. Larger devices need their own supply.

Form the Orin, U6 draws 138mA typical and 260mA worst case from 5V. On 3.3V the two CDIPs draw about 140MA each at full load, about 0. 3 in total. Both rails are well below 1A.

4.3 Interface

The BOB plugs onto the Orin 40-pin header (I2C5, SPI1, UART1, CAN0 on pin 29/31, PWM01 and two GPIO, 3.3V logic) and gets 3.3V and 5V from it. Pin 33 is used as GPIO, so the second Orin CAN bus is not available. Outputs are J6 (I2C), J10 (SPI), J9 (UART + GPIO) with isolated 3.3V, and J7 (CANH, CANL, PWM). It does not deliver power except the small isolated 3.3V rails.

Orin Interface

1 Design overview

The Orin Interface is the lowest PCB in the control house, mounted above the Jetson AGX Orin. It takes protected battery power in and supplies the Orin and the two Ethernet switches (BotBlox GigaBlox and Rugged GigaBlox). It also routes gigabit Ethernet from the switches to the three cameras, the acoustics unit and optional two non-PoE devices and supplies Power over Ethernet (PoE) to the cameras and acoustics, since the switches cannot. The switches, the Orin, the connected devices and the cables are outside the scope.

2 Design inputs and constraints

2.1 Key requirements

All shared requirements under section 4 in the Electrical System Specifications apply to the Orin Interface [SPEC, Sec. 4]. The most relevant are listed in table **??**

2.2 Rationale

The cameras and acoustics need gigabit data and get power over the same cable, which keeps the number of penetrators down. They are standard PoE devices, so the board must acts as an 802.3at PSE. The board is the entry point for battery power to the Orin and switches, so a short, reverse connection or over-voltage must be stopped here before it reaches expensive hardware.

3 System architecture

Battery power enter at J1 and goes through the eFuse to the protected rail `BATT_P` (figure 5). From there is goes straight to the switches and the Orin, to U4 (3.3V for the PoE controller) and to the boost converter, which makes 47.5V for the PoE controller. Ethernet from the switches passes through four PoE injection blocks, where the PoE controller adds power, and out to the cameras and acoustics. The non-PoE devices is a direct data link from Switch B.

4 Detailed design

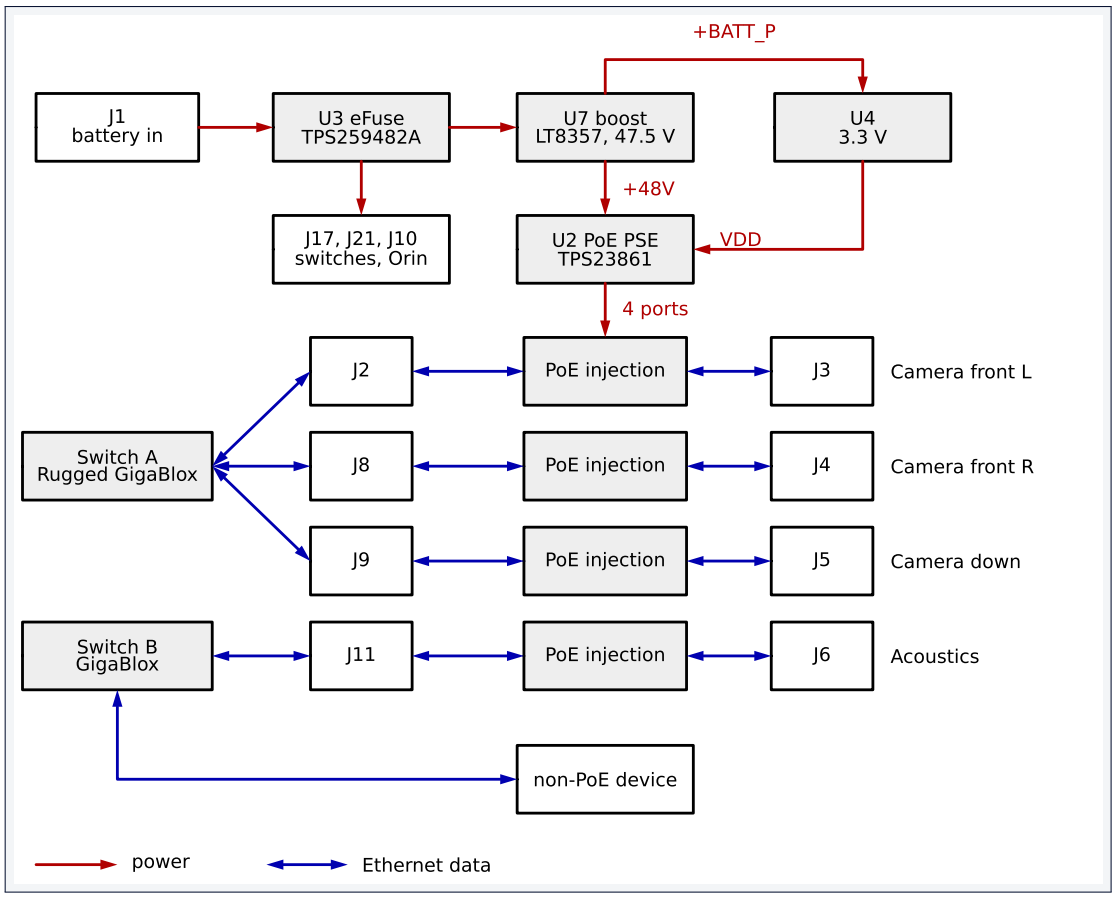


Figure 5: Orin Interface architecture.

4.0.1 eFuse

The TPS259482A (U3) is an 8A, 3.5-23V eFuse with Adjustable UVLO, OVLO, current limit, inrush control and reverse-current blocking in one small package. The A version retries automatically after a fault. RCBCTRL is left open, which keeps reverse blocking on.

4.0.2 Boost converter

The LT8357 (U7) is a boost controller with an external MOSFET, chosen because it handles a high step-up ratio at tens of watts , and the datasheet has a 48V reference design we based ours on. It runs at 350 khZ (RT = 95.3$k\Omega$) with Burst Mode at light load. The boost has no output short-circuit protection of its won. The ports are protected by the PoE controller and fuses F1-F4.

4.0.3 PoE controller

The TPS23861 (U2) is a four port 802.3at PSE controller. It was chosen because it detects, classifies and powers each port on its own without a microcontroller. SCL and SDAI are tied low and SDAO left open for auto mode. Per port parts (47Ω, 22.1Ω, $o.255\Omega$ sense resistor, $0.1\mu F$) are the datasheet values. In auto mode it sets the port limit to 374mA or 645mA.

4.1 Calculations and margins

eFuse. The turn-on and cut-off voltages are set with resistor dividers:

$$V_{\mathrm{UVLO}} = 1.2\,\mathrm{V} \cdot \frac{470\,\mathrm{k\Omega} + 45.3\,\mathrm{k\Omega}}{45.3\,\mathrm{k\Omega}} = 13.7\,\mathrm{V} \tag{2}$$

$$V_{\mathrm{OVLO}} = 1.2\,\mathrm{V} \cdot \frac{470\,\mathrm{k\Omega} + 31.2\,\mathrm{k\Omega}}{31.2\,\mathrm{k\Omega}} = 19.3\,\mathrm{V} \tag{3}$$

The board turns on at 13.7 V and turns off again at 12.5 V, which protects the battery from going too low. Anything above 19.3 V is blocked.

The current limit is set by R_{ILM}:

$$I_{\mathrm{LIM}} = \frac{4834}{R_{\mathrm{ILM}}} = \frac{4834}{576\,\Omega} = 8.4\,\mathrm{A} \tag{4}$$

Boost converter. The output voltage is set by the feedback divider:

$$V_{\mathrm{OUT}} = 1\,\mathrm{V} \cdot \frac{1\,\mathrm{M\Omega} + 21.5\,\mathrm{k\Omega}}{21.5\,\mathrm{k\Omega}} = 47.5\,\mathrm{V} \tag{5}$$

This is within the 44–57 V the PoE controller needs. The sense resistor limits the peak current to 60 mV/10 mΩ = 6 A. At a low battery (12.5 V) this gives about 45–64 W of output power.

PoE budget. One port can draw up to 47.5 V · 645 mA = 30.6 W. Four ports at full power would be more than the boost converter can deliver, so the cameras and acoustics together must stay below about 45 W

4.2 Interface

Power comes in on J1 from the BMS (4S battery). The board delivers `BATT_P` to Switch A (J21), Switch (J17) and the Orin (barrel jack J10), all sharing the 8.4A eFuse limit. Ethernet comes in fro mthe switches on J2, J8, J9 and J11, and goes out with PoE on J3 (camera front left), J4 (camera front right), J5 (camera down) and J6 (acoustics). The board has no CAN or other control interface.

Power Interface

1 Design overview

The power-interface PCB will act as the main interface between the connector plate and the rest of the control-house. It's main job is to facilitate connections to various outside systems like the actuator houses, lights and gripper. It will also be responsible for monitoring these systems in different ways like their current draw and CAN-bus activity, which it will log to an external storage. The power-interface PCB will also be responsible for the killswitch, as it already distributes other signals throughout the AUV.

2 Design inputs and constraints

2.1 Key requirements

All shared requirements under section 4,5 and 6 in the Electrical System Specifications apply to the Power Interface PCB. All requirements under section 11 also apply to the PCB as well as subsections 10.1 and 10.2 under section 10 [SPEC, Sec. 4, 5, 6, 10.1–10.2, 11]. The most relevant of these can be seen in table 5.

2.2 Rationale

Define the requirements and constraints in more detail. Explain why the requirement is set and why it is set to the spesific value if applicable.

Rationale for all the Power Interface rewuirements can be found in the Technical spesifications [SPEC, Sec. 4, 5, 6, 10.1–10.2, 11], but the rationale for those listed in 5 are as follows:

6.1	This is just smaller than diameter of the electrical housing we receive from our partners. To fit as many PCBs as we need into the control house, the PCBs should fit this diameter, as that allows us to stack them like oreos. We also need a space buffer for the 3D printed coponents that will hold the PCBs.
11.3	It is nice to know how much current each of the different circuitry connected to the control house draws, and monitoring from each of the actuator houses will allow us to see if one of the batteries is being drained more than the other.
11.5	This will allow us to review what happened on the CAN-bus during a tetherless run, like Robosub. This can then be used to troubleshoot if there were issues during the tehtherless run.
11.6	We need to make sure we don't fry the PCB and this will protect our components such that the consequence of short-circuits or human errors deacreses.

Table 3: Power Interface key design-driving requirements

Req-nr	**Requirement / target**	**Design impact**
REQ-6.1	The PCB shall fit entirely within a circle with a diameter of 130 mm	Components must be chosen so that they'll fit
REQ-11.3	The PCB must be capable of individually monitoring the current draw of itself, the control-house and the current draw from each of the actuator houses	Need circuitry to measure the current at each of these positions
REQ-11.5	All CAN-messages must be logged to an easily accessible non volatile storage medium like an SD-card	The PCB must include a connector for some kind of external storage
REQ-11.6	Actuator-house power inputs must have reverse-polarity protection. All power outputs must have short circuit protection	The PCB needs to have some kind of e-fuse or similar protection that meets the requirement on each power output
REQ-11.7	The power-interface must provide power, CAN-bus and a killswitch signal to the gripper	The design needs to make sure it can handle the large current draw of the gripper
REQ-11.10	The killswitch must be connected such that the killswitch specifications are met	Must make sure that all actuators on the AUV receive killswitch and GND signal
REQ-11.12	The actuator house power inputs must have ideal diodes	Both the power going to the control house and the gripper must be power OR-ed, with ideal diodes that can handle the high current

11.7 The gripper needs all of these signals and it is connected to the control house. The power interface is the most natural place to provide these.

11.10 Robosub requires a way to shut off all mechanical actuators. Doing this hardwired ensures that no software bugs interfere with this requirement. As the killswitch is attached to the control house, the power interface is the most natural PCB to send out this signal.

11.12 We don't want the batteries to be able to power each other.

3 System architecture

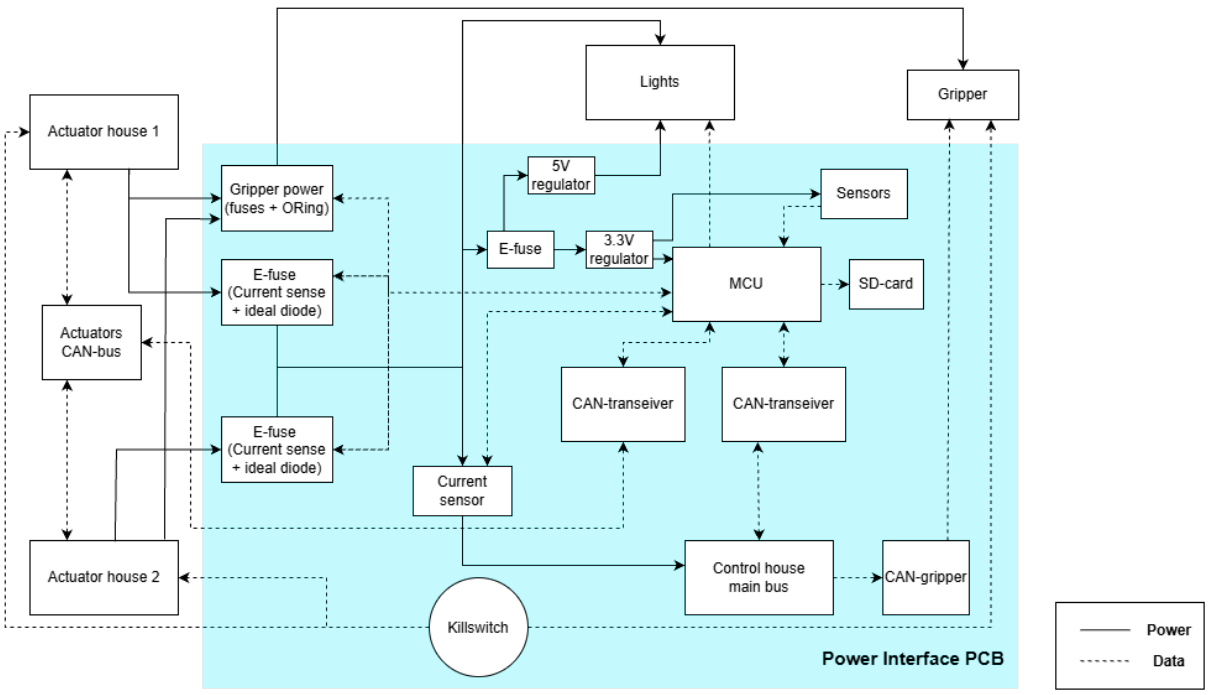


Figure 6: Power Interface architecture overview

The system architecture can be seen in figure 9.

Starting with the power lines, we can see that the PCB takes in power from the actuator houses. Here we see the first major design choise, being to separate the power going to the gripper from the rest of the control house. This is done to satisfy REQ-11.7 while making sure that a short or similar errors with the high current going to the gripper won't shut down the power going to the rest of the control house. In the E-fuse and gripper power blocks connected to the actuator houses we also cover REQ-11.3, REQ-11.6 and REQ-11.12. The E-fuses being used to protect the PCB at the power outputs, also give us current monitoring of the current coming from each of the actuator houses. On these inputs there has also been placed ideal diodes.

The battery voltage is then delivered to the camera lights and control house main power bus, which has a hall-effect current sensor monitoring the current draw of the rest of the control house, covering the rest of REQ-11.3. The battery voltage also goes through the standard E-fuse for the MCU [SPEC, Sec. 4.7]. This current then goes through the 5V regulator to the indicator lights and 3.3V regulator for the MCU and attached sensors and ICs.

On the signal side we have to CAN-buses connected to the MCU through each their CAN-tranceiver. The MCU will then be used to facilitate communication over the two buses, and it is

attached to a microSD connector to save the messages, as per REQ-11.5. The MCU also receives data from the different sensor on the PCB (current and temp) while sending out data to the idicator and camera lights. Lastly the PCB takes in an external killswitch signals and transmists that to the actuator houses and the gripper per REQ-11.10.

4 Detailed design

4.1 Microcontroller

The STM32C542CCT6 handles sensor monitoring, communication, SD-logging and idicators following the shared and PCB spesific requirements. Sensor communication is done through I2C and analog pins, while the indicator lights and SD-card uses SPI. The camera lights are interfaced using PWM, CAN-tranceivers are connected to FDCan pins and the MCU also provides us a UART and SWD interface for debugging. Pin functions can be seen in the cited datasheet. [SPEC, Sec. 4, 11] [PI-1]

4.2 MCU E-fuse

To protect the voltage regulators supplying the MCU and other low voltage ICs on the PCB we use a TPS26400PWPR [PI-2] as per the shared requirements [SPEC, Sec. 4.7–4.9]. The OVLO and UVLO are set following the spesifications, while current limitng on this PCB is set to 600 mA. This is due to each of the regulators pulling 0.5 A at their respective voltage resulting in a total current drawn from the 15 V e-fuse supply at:

$$\frac{0.5\,\mathrm{A}\cdot 3.3\,\mathrm{V} + 0.5\,\mathrm{A}\cdot 5\,\mathrm{V}}{15\,\mathrm{V}} \approx 280\,\mathrm{mA} \tag{6}$$

As the power regulators may draw more current in short burts, the expected current is about doubled and that is set as the current limit. [PI-2]

4.3 Main control house power bus E-fuses

Two TPS259480LYWP are used to protect and power-OR the main control house power bus. These work as ideal diodes and current sensors, while having all required protection except for reverse polarity. Therefore there is an added reverse polarity protection circuit on each power input. OVLO is 19.4V and UVLO is 12V as these are the values work well for the BM, and current limit is set to 8.6A on each E-fuse giving the control house an available current of 17.2 A. [PI-3]

4.4 Gripper E-fuse

As the gripper can pull up to 20A (this is what our wires are rated for) we want its power input to have its own bus. For the high current we use an NCP81296MNTXG, as it can handle a high current. Current limit is set to 21.8A, as the wires can handle a bit more in the wate. The current limit is set using $R_{CLREF} = \frac{V_{CLREF}}{I_{CL}} = \frac{R_{CS}(I_{LIM}\cdot 10\mu\mathrm{A/A}+5\mu\mathrm{A})}{10\mu\mathrm{A}}$, and the R_{CS} value is set to a standard value of 2kΩ. More information can be found in the datasheet. [PI-4]

4.5 CAN-tranceivers

Required CAN-interfacing is handled by the ADM3057EBRWZ and TCAN3414DR. The first is isolated and handles communication over the actuator house CAN-bus, using a 5 V supply and 3.3 V MCU logic, isolating both supply and GND. Here the datasheet recommends splitting GND into an isolated GND and PCB GND. This was diregarded after advice that splitting GND could introduce more problems into the PCB. The second tranceiver is the standard unisolated tranceiver for communication on the control house CAN-bus. [PI-5] 6

4.6 Control house power bus current sensor

The TMCS1100A3QDR handles current monitoring on the power bus going to the contorl house. It is a galvanically isolated hall effect sensor with $200\frac{\text{mA}}{\text{V}}$ precision, using the 3.3 V source. This allows for current sensing up to 16.5 A using an ADC on the MCU, leaving a small gap from 16.5 to 17.2 A (The E-fuse current limit) where we'll only sense 16.5A. We can't change the source to 5 V as the MCU has a 3.3 V logic level, and we've determined that the tradeoff in resolution (50% reduction in resolution) when going to the next precision level is not worth it, as the drone shouldn't operate in this current level much, except for spikes. Backwards current can't be measured either, but that should not be possible with the reverse polarity and short curcuit protection. [PI-7]

4.7 SD-card

The micro-SD card is connected using an 693071020811 connector and is interfaced using half-duplex SPI, as we only need to write to it. It runs using 3.3V and has pull-ups on each of its datalines. All messages sent over the CAN-bus will be stored on this MicroSD card, which will be accesible between tests. [PI-8]

Sensor Interface

1 Design overview

The Sensor Interface PCB is the main interface between the vehicle's navigation sensors and the Nvidia Orin AGX. It powers, configures and reads the sensors, timestamps their measurements and forwards the data to the Orin over Ethernet. The connected sensors are one STIM300 IMU[SI-12], a Nortek Nucleus 1000 DVL[SI-13], a BAR30 pressure sensor[SI-14] and a magnetometer from a separate PCB[See "Magnetometer PCB"]. The board handles sensor communication and timing, while navigation and sensor fusion are performed on the Orin. [SPEC, Sec. 12.1, 12.7]

2 Design inputs and constraints

The design follows the shared electrical requirements and the Sensor-interface requirements in Section 12 of the specification. The main constraints are sensor power protection, interface isolation and accurate measurement timestamps. The board must also fit the common control-house outline and provide the shared power and CAN connection. Würth components are used extensively for passives, connectors and power modules, following the component preferences in the specification.[SPEC, Sec. 4.1, 5-6, 12]

2.1 Key requirements

Table 4: Sensor Interface design-driving requirements

Source ID	Requirement / target	Design impact
REQ-4.3	Sensor Interface may use STM32H753VIT6.	Allows the MCU needed for sensor interfaces, Ethernet and timing functions.
REQ-4.7–4.9	Protected power inputs and current monitoring.	Requires eFuses and a current measurement path to the MCU.
REQ-5.3	Shared 24-pin control-house connector.	Determines the power, CAN and inter-board connections.
REQ-6.1–6.4	Common outline, cutouts and component keepout zones.	Constrains PCB layout and connector placement.
REQ-12.3–12.4	Power the sensors and protect their supply lines.	Requires suitable power distribution, short-circuit protection and overvoltage protection.
REQ-12.5–12.6	Isolate sensor digital signals and use relevant auxiliary signals.	Requires isolated interfaces and connections for triggering and timing. I2C isolation is not included in the current design.
REQ-12.7–12.9	Timestamp measurements and synchronize the MCU with the Orin.	Requires hardware timing and a PWM input with input capture.
REQ-12.10	Dedicated CAN bus for the magnetometer.	Requires a second CAN interface separate from the vehicle bus.

2.2 Rationale

The sensors are expensive and connect to several different power and signal interfaces. Protection is therefore needed on both the supply and communication connections. Accurate timestamps are also important: the Orin needs to know when a measurement was taken, rather than only when it reached the MCU. Trigger and timing signals are included to support this. [SPEC, Sec. 12.4, 12.10]

3 System architecture

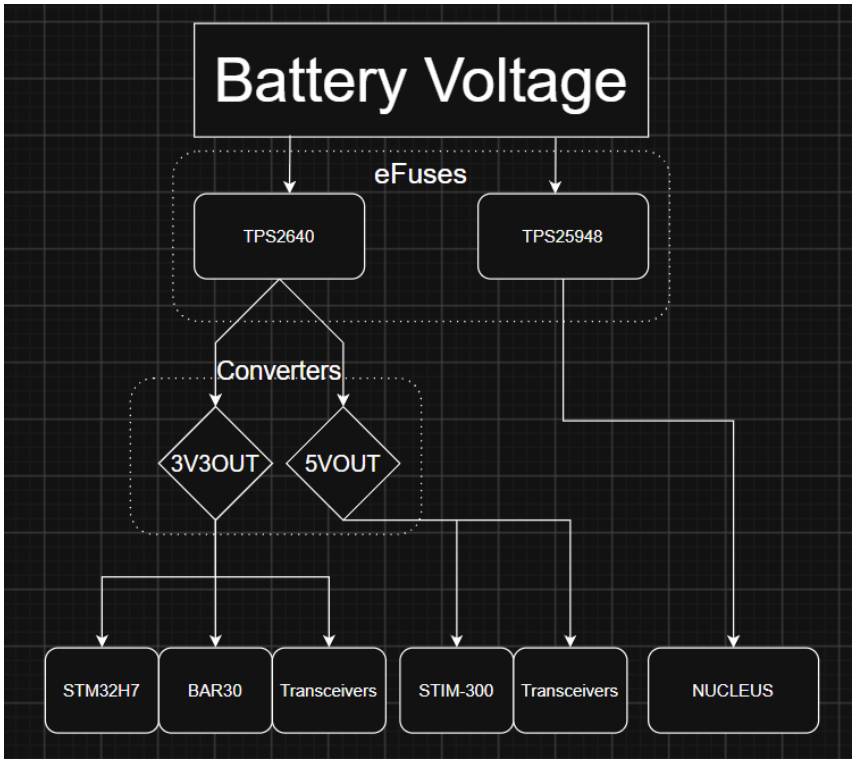


Figure 7: Sensor Interface power distribution, showing the protected supply branches for the Nucleus 1000 and the 3.3 V and 5 V regulators.

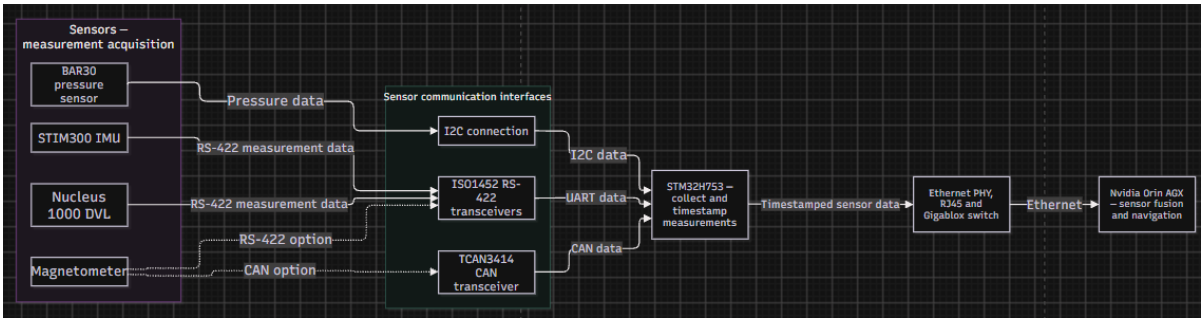


Figure 8: Sensor data flow from the sensors through their communication interfaces to the STM32H753, which timestamps and forwards the measurements to the Orin over Ethernet.

Power enters through the shared control-house connector and is divided into two protected branches. A TPS26400PWPR[SI-2] supplies the inputs of the 3.3 V[SI-5] and 5 V[SI-4] converters, while a separate eFuse supplies the Nucleus 1000 from the vehicle power bus. The BAR30 communicates directly with the MCU over I2C, without galvanic isolation in the current implementation.

Three RS-422 interfaces are provided for the STIM300, Nucleus 1000 and a possible RS-422 magnetometer. The pressure sensor uses I2C. Two CAN interfaces separate communication with the vehicle from the dedicated magnetometer bus. Sensor measurements are forwarded to the Orin through the Ethernet PHY and the vehicle's Gigablox Ethernet switch. A timing input from the Orin supports clock synchronization, while sensor trigger and timing connections support measurement timestamping. [SPEC, Sec. 12.2, 12.7-12.10]

4 Detailed design

4.1 Microcontroller

The STM32H753VIT6[SI-1] is mounted directly on the PCB in a 100-pin LQFP package. It handles the sensor interfaces, Ethernet communication and synchronization with the Orin. This MCU is explicitly permitted for the Sensor Interface in the specification. The main challenge is allocating the available pins so that UART, CAN, RMII, I2C and the required timer channels can operate together. [SPEC, Sec. 4.3]

4.2 Voltage regulators

The board uses the Würth 173020536 module for the 5 V supply and the 173950336 module for the 3.3 V supply. The 5 V rail supplies the STIM300 and the associated loads requiring this voltage, while 3.3 V supplies the MCU and local logic. The 173020536 provides a rated output current of 2 A, leaving capacity beyond the discussed 0.4 A STIM300 load for other connected electronics. The 3.3 V module follows the shared regulator requirement. [SPEC, Sec. 4.5]

4.3 Power protection

The TPS26400PWPR protects the input supply to both the 3.3 V and 5 V converters. It therefore protects the power branch supplying the MCU, local interface electronics and sensors powered from these regulated rails. Its current limit must account for the combined input current of both converters, including conversion losses and startup current. This component follows the shared eFuse requirement. [SPEC, Sec. 4.7, 4.9]

The Nucleus 1000 is supplied from the vehicle power bus through a separate eFuse[SI-3], rather than through the 3.3 V or 5 V converter. This provides a dedicated protection path for the DVL. Its protection thresholds must be matched to the sensor's supply requirements. [SPEC, Sec. 12.3-12.4]

The protection thresholds must suit the loads supplied by each branch. Current monitoring is required for the board and connected devices, while fault signals allow the MCU to report protection events. Final current limits, startup timing and voltage thresholds must be documented from the schematic calculations. [SPEC, Sec. 4.7-4.9, 12.4]

4.4 RS-422 interfaces

Three ISO1452BDW[SI-6] isolated full-duplex transceivers are included for the STIM300, Nucleus 1000 and optional RS-422 magnetometer. Each transceiver converts between the MCU's UART signals and separate differential transmit and receive pairs. Integrating the signal isolation into the transceiver reduces the need for separate isolators on these serial communication paths.

The isolation boundary must be maintained by the power and ground arrangement on both sides of the transceiver. Termination must match the receiving end of each link. Sensor trigger and auxiliary signals require their own suitable interface circuitry. [SPEC, Sec. 12.2, 12.5-12.6]

4.5 Magnetometer and CAN interfaces

Two TCAN3414DR[SI-7] transceivers provide separate CAN interfaces for the vehicle network and the magnetometer. A dedicated magnetometer bus follows the specification and keeps its communication separate from the main vehicle bus.

The BOM includes two 120 Ω termination resistors and a termination switch. This allows the termination arrangement to be adapted to the intended bus topology and test setup. The TCAN3414DR does not itself provide galvanic isolation; Section 12.10 explicitly states that isolation is not required for the dedicated magnetometer CAN connection.

4.6 Sensor timing and synchronization

The MCU receives the STIM300 Time of Validity signal and provides the required sensor trigger outputs through a two channel digital isolator [SI-8]. A separate PWM input from the Orin is connected to a timer channel with input-capture capability. These connections support timestamping at the measurement time and synchronization between the MCU and Orin clocks. [SPEC, Sec. 12.6-12.9]

Hardware timers are used to capture timing edges and generate trigger signals. The final firmware must account for each sensor's timing behaviour when assigning timestamps, since receiving a data packet does not necessarily indicate when its measurement was taken.

4.7 Pressure sensor

The Blue Robotics BAR30 measures external water pressure for depth estimation and communicates with the STM32 over I2C. The Sensor Interface PCB provides power and the SDA and SCL connections, with pull-up resistors on the bus. [SPEC, Sec. 12.1- 12.3]

4.8 Ethernet

The LAN8742AI-CZ-TR PHY[SI-9] connects the MCU's RMII interface to Ethernet. The cable connection uses the Würth 74990111217 RJ45[SI-10] connector with integrated magnetics, as listed in the BOM. Ethernet carries the timestamped sensor data to the Orin through the Gigablox switch. [SPEC, Sec. 12.1- 12.2]

The PHY circuit includes the Würth 830055299[SI-11] crystal, configuration resistors and supply decoupling. A separate reset connection allows the MCU to control PHY initialization.

4.9 Debugging and ESD protection

The board includes a 629722000214 USB-C connector[SI-15], a 10-pin SWD header and a UART debug header. These provide access for programming, firmware development and troubleshooting. Programmable red, yellow and green LEDs provide local status indication, while test points expose supply rails and communication signals. [SPEC, Sec. 4.12, 4.14-4.19]

The BOM includes the specified NSR0530HT1G Schottky diode for the USB power arrangement and several Würth TVS devices for exposed connections. These include the 824501261 specified for battery-voltage inputs. Protection placement must cover the connectors and exposed test points identified in the specification. [SPEC, Sec. 4.13, 4.22]

Torpedo and Markerdropper

1 Design overview

The torpedo and marker dropper PCBs control the actuators in their respective subsystems, acting on commands that the central computer distributes over the CAN bus. One PCB is mounted on top of each actuator housing: one for the torpedo launcher and one for the marker dropper. In addition to actuator control, each PCB carries a telemetry sensor and is connected to indicator lights.

2 Design inputs and constraints

Table 5: Power Interface key design-driving requirements

Req-nr	Requirement / target	Design impact
REQ-7.1	All PCBs in the actuator-house must fit in the actuator house.	Components must be chosen so that they'll fit
REQ-16.2	PCB needs to be equiped with pogopads	Power flow is designed around this central power bus
REQ-16.3	Input power must have reverse polarity and UVLO functionality for the servos power input	Need circuitry to adhere to the requirement
REQ-16.4	Current overload protection and short circuit protection is required for the power lines going to the servo	Need circuitry to adhere to the requirement
REQ-16.5	Must be capable of measuring the servos current draw.	Need circuitry conecting the efuses to the mcu to messure current draw
REQ-16.6	This PCB must be equipped with a BME280 pressure, temperature and humidity sensor.	Need an BME280 sensor connected to the MCU
REQ-16.7	Active killswitch must kill all power to the servos	Need to make sure killswitch implementation is hardware controlled

2.1 Rationale

7.1 The PCB must fit within 70mm, as that is the inner dimensions of the actuator-house. The PCB will also have 4 indents around the edges used for screwing it firmly to the housing, so the circuits will need distance from the edges.

16.2 The pcb is among other connectors using pogopads. Pogopads allows the rest of the actuator house to be easily removed, modified and slid back inn place without fiddeling with wires.

16.3 It is important to protect the PCB from under voltage, and reverse polarity. Adding the extra circuitry minimizes the chances of faults.

16.4 It is important to protect the PCB from overloading, and short circuiting. Adding the necessary protection is a cheap way to negate the chances of frying the board.

16.5 It is important to measure the servos current draw to estimate the servos outputted

torque and promptly inform the user of any anomalies.

16.6 This PCB must be equipped with a BME280 pressure, temperature and humidity sensor. The measurements must be posted on to the CAN-bus at 1 Hz. The I2C protocol is chosen to save space, and to standardize between boards.

16.7 Robosub requires a way to shut off all mechanical actuators. Doing this hardwired ensures that no software bugs interfere with this requirement. This insures that the actuators will not get power when the Killswitch is active.

3 System architecture

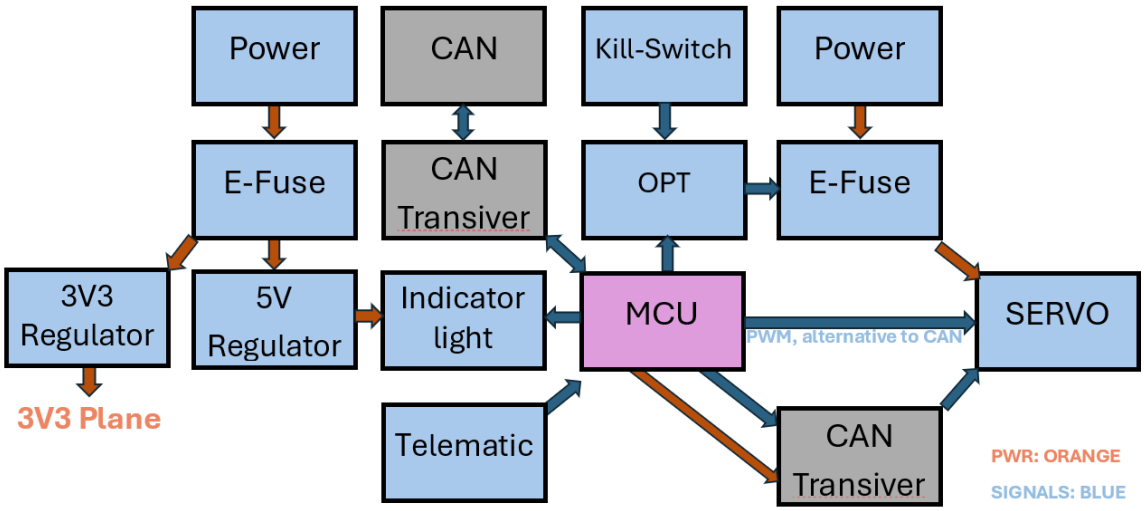


Figure 9: Torpedo / Marker dropper architecture overview

Power enters through the pogo pads and exits through the top of the actuator house. The PCB taps the power bus and routes it through both e-fuses. It supports two types of servo: a CAN servo with a high operating voltage, and a PWM servo with a lower operating voltage of 6.7 V.

The PCB uses two e-fuses. The larger one limits the current going to the servos, and the smaller one sits upstream of the regulators.

The vehicle battery voltage (V-BAT) is 16 V, so it is stepped down by a 3.3 V and a 5.0 V buck converter. The board is a four-layer PCB, with one layer dedicated to a 3.3 V plane. The BME280 sensor, MCU, CAN transceivers and status LEDs, among other components, are powered from this plane. The 5 V rail only supplies the external indicator light, as it is the only component that requires 5 V. A power module after the larger e-fuse steps the voltage down to 6.7 V, as PWM servos do not support V-BAT.

An STM32 microcontroller (MCU) acts as the brain of the PCB. It routes information between the components and sends and receives CAN messages.

The PCB uses two separate CAN buses. The first connects to the Orin, from which it receives commands and to which it sends status messages and sensor data. The second is a local bus used to communicate with the actuator. It is only used if a CAN-compatible actuator is connected, as opposed to a PWM servo. CAN transceivers convert between the MCU's CAN signals and the differential bus signals.

The PCB implements a hardware killswitch using an optocoupler. This ensures the killswitch works without any software involvement.

The PCB is equipped with a BME280 pressure, temperature and humidity sensor. It is powered from the 3.3 V plane, continuously monitors the air inside the actuator housing, and sends the data to the MCU.

4 Detailed design

4.1 Actuators

The PCB's main task is controlling up to two servos, and it supports two servo types. The preferred servo is the Blue Trail Engineering Underwater Servo SER-213X, because of its high operating voltage (32 V) and CAN bus support. The other supported servo is the Blue Trail Engineering Underwater Servo SER-2020, which uses approximately 6.7 V and PWM signals. They are connected through the 91311400102 and 691382000003 connectors.

4.2 Microcontroller

The STM32C542CC microcontroller (MCU) acts as the brain of the PCB. It routes information between the components and sends and receives CAN messages. The 44-pin package was chosen to minimize board space. The MCU is connected to UART, SWD and a USB-C 2.0 port for debugging. It also drives three programmable status LEDs (red, yellow and green, 150060RS55040) that indicate faults and irregularities.

4.3 Servo E-fuse

The PCB uses two e-fuses. The larger one (NCP81296) limits the current going into the servos. This e-fuse does not incorporate reverse polarity protection, so a dedicated circuit between the power connectors and the e-fuse's VIN pin provides it, as required by [SPEC, Sec. 16.3]. The e-fuse has several unsoldered 0 Ω resistors after its current output. They are only soldered in place once the e-fuse has been tested and deemed operational, which ensures that a faulty e-fuse cannot damage the rest of the board.

4.4 Lower-current E-fuse

A smaller e-fuse (TPS2640) is placed upstream of the regulators. It has built-in UVLO and reverse polarity protection, as required by [SPEC, Sec. 16.3]. It uses 0 Ω resistors in the same way as the servo e-fuse.

4.5 Converters

The vehicle battery voltage (V-BAT) is 16 V, so it is stepped down to 3.3 V and 5.0 V by two buck converters (173950336 and 173950536). A power module (171032401) placed after the larger e-fuse steps the voltage down to 6.7 V, as PWM servos do not support V-BAT.

4.6 CAN Transceivers

The CAN transceivers (TCAN3414DR) convert between the differential CANH/CANL signals and the MCU's TX/RX signals. A 120 Ω termination resistor is connected between CANH and CANL on the local servo bus. The first CAN bus has one as well, but it is connected through a switch so that it is only active during testing.

4.7 Telemetry Sensor

The PCB is equipped with a BME280 combined pressure, temperature and humidity sensor, as required by [SPEC, Sec. 16.6]. The sensor is powered from the 3.3 V plane and sends its data to the MCU over I2C. The data is forwarded to the Orin at 1 Hz.

4.8 Killswitch

An optocoupler (140356145200) is used for the killswitch to ensure the mechanism works reliably without software. The killswitch circuit can also be controlled in software, through an MCU output driving a MOSFET.

Magnetometer PCB

1 Design overview

The magnetometer PCB gives the AUV an absolute heading reference. The STIM-300 IMU can track changes in heading by integrating its gyro rates, but it cannot determine absolute heading on its own, and the estimate drifts over time due to gyro bias. The magnetometer provides the absolute reference needed to initialise the heading and correct this drift [SPEC, Sec. 12.1].

Magnetometers are sensitive to fields from motors and electronics, and the AUV is small, so the sensor is placed on its own board. This lets us move it around in the vehicle to find the position with the least disturbance.

The board carries an MMC5983MA three-axis magnetometer, which communicates over SPI. The board is connected to the sensor-interface PCB with a cable of roughly 40–50 cm, which is too long for SPI. An STM32C542 on the board therefore reads the sensor and forwards the measurements over a dedicated CAN bus to the sensor interface.

2 Design inputs and constraints

2.1 Key requirements

Most requirements that apply to this board are the shared requirements in Section 4 of the specification sheet. The board-specific requirement is the dedicated CAN bus in [SPEC, Sec. 12.10]. The most relevant are listed in Table 6.

Table 6: Magnetometer PCB design-driving requirements

Source ID	Requirement / target	Design impact
REQ-12.1	Magnetometer for heading reference	MMC5983MA, 18-bit, ±8 G, SPI
REQ-12.10	Communicate with the sensor interface over a separate, dedicated CAN bus	Local SPI-to-CAN conversion with an MCU and CAN transceiver
REQ-4.3, 4.11	STM32C542CCT6, FDCAN2 on PB5/PB6	Shared MCU; CAN bootloader available
REQ-4.10	TCAN3414DR or ADM3057 transceiver	Non-isolated TCAN3414DR, since the bus does not need isolation
REQ-4.4	Accept the battery voltage directly	Buck module rated 3.5–36 V input
REQ-4.12, 4.13	USB-C 629722000214, NSR0530HT1G from VBUS	Board can be powered and debugged from USB alone
REQ-4.14, 4.17–4.20	UART, SWD with SWO, reset button, 48 MHz HSE oscillator	UART pins, ST-LINK header, reset switch, 830207659609
REQ-4.22, 4.23	TVS on all connectors, 0 Ω on supply outputs	ESD arrays on every connector; R1 on the 3.3 V output

2.2 Rationale

CAN was chosen over extending SPI because it is differential and tolerates long cables and noise much better, and because the team already has CAN firmware and tooling. The bus is dedicated to this sensor and does not need to be isolated, which allows the simpler TCAN3414DR and reuse of the firmware and schematics from the other boards.

Using the shared STM32C542 rather than a dedicated bridge IC keeps spare parts and the CAN bootloader common with the other boards. It also lets the board configure the sensor, run its SET/RESET sequence and pace the measurements locally, so the sensor interface only receives finished samples.

The MMC5983MA was chosen for its low noise (0.4 mG RMS at the lowest bandwidth setting) [MP-6]. Its built-in SET/RESET function removes offset caused by temperature drift and residual magnetisation, which matters in a vehicle with thruster magnets close by. It has a 10 MHz SPI interface and a data-ready interrupt, and comes in a small 3×3 mm package.

2.3 Deviations from the shared requirements

Since this is a small sensor board powered through the sensor interface, some shared requirements are not met as written. These deviations are listed in Table 7.

Table 7: Deviations from the shared requirements

Source ID	Shared requirement	Implementation and justification
REQ-4.7, 4.8	TPS26400PWPR eFuse with shared UVLO/OVLO on the power input	No eFuse on the board, to keep it small. V-batt is supplied through an eFuse on the sensor-interface PCB, which protects the cable and this board. Locally, a TVS diode and a P-channel MOSFET give overvoltage and reverse-polarity protection.
REQ-4.5	Würth 173950336 3.3 V regulator	Due to size constraints, a Würth 171930601 (VDMM) is used instead. It is 5×2.5 mm and its 300 mA rating is well above the board's load, which suits the small board.
REQ-4.15	Blue LED on each rail, red/green eFuse fault LED, MCU-controlled red, yellow and green LEDs	MCU drives red, yellow and green LEDs, and a blue LED shows that the 3.3 V rail is present. There is no eFuse on the board, so there is no eFuse fault LED.

3 System architecture

The architecture is shown in Figure 10. Battery voltage, CAN and a PWM line arrive from the sensor interface on a single connector. The battery voltage passes the input protection and is converted to 3.3 V, which supplies every IC on the board. USB is an alternative supply joined in after the input protection. The microcontroller reads the magnetometer over SPI when its data-ready interrupt fires, and sends each sample on FDCAN2 through the CAN transceiver. Debug access is through USB, the uart pins, the SWD header and an SPI test header.

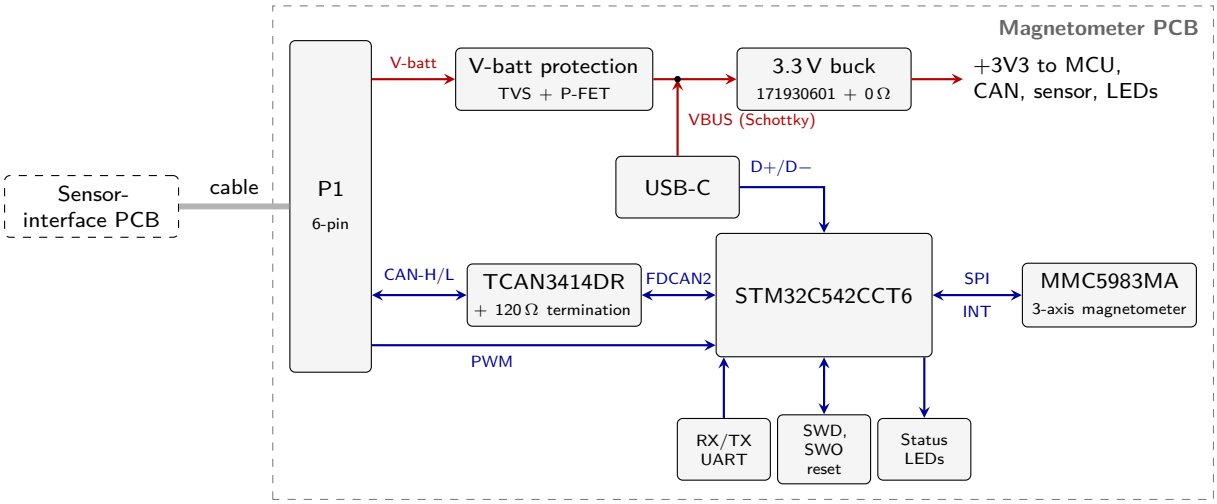


Figure 10: Magnetometer PCB architecture. Red: power, blue: data and control.

4 Detailed design

4.1 Magnetometer

The MMC5983MA is a three-axis magnetometer with ±8 G range and 18-bit resolution [MP-6]. It runs at 50 Hz, the lowest bandwidth setting, which gives the lowest noise of 0.4 mG RMS. It is

connected to SPI1, and its data-ready output goes to the MCU so each sample is read as soon as it is ready. The CAP pin has the 10 μF capacitor required for the sensor's SET/RESET function, which removes offset caused by temperature drift and strong nearby fields.

4.2 Microcontroller

The STM32C542CCT6 is the shared MCU [MP-1], with FDCAN2 on PB5/PB6 so the built-in CAN bootloader can be used [SPEC, Sec. 4.3, 4.11]. Since the board is on a dedicated CAN bus, updating the firmware over CAN would require the sensor-interface firmware to forward the update from the Orin. It uses the shared 48 MHz oscillator as HSE [MP-4] and a 32.768 kHz oscillator as LSE [MP-5].

A PWM signal from the sensor interface is connected to TIM2 on PA8. The sensor interface uses this signal to synchronise all sensor data [SPEC, Sec. 12.7–12.9]. On this board, TIM2 captures the PWM edges so each magnetometer sample can be related to the shared time reference before it is sent on CAN. This removes the uncertainty from the variable CAN delay. TIM2 was chosen because it is a 32-bit timer, so it can count at high resolution for a long time without overflowing.

4.3 CAN interface

The TCAN3414DR runs from 3.3 V, the same as the MCU [MP-3]. The board is one end of the bus, so it has a 120 Ω termination resistor. CAN-H and CAN-L are protected by a TVS array at the connector.

4.4 Power

V-batt is protected by the standard 824501261 TVS diode [SPEC, Sec. 4.22] and a P-channel MOSFET for reverse polarity. A 171930601 buck module makes the 3.3 V rail [MP-2]. It is small (5×2.5 mm), delivers up to 300 mA and has built-in current limiting and undervoltage lockout. USB VBUS is connected through an NSR0530HT1G Schottky diode, so the board can run from USB without backfeeding the PC [SPEC, Sec. 4.13]. A 0 Ω resistor on the regulator output is fitted after the rail has been tested [SPEC, Sec. 4.23].

4.5 Calculations and margins

Output voltage. The output voltage is set by the feedback divider, with the module's internal reference $V_{\mathrm{REF}} = 0.75\,\mathrm{V}$ [MP-2]:

$$V_{\mathrm{OUT}} = V_{\mathrm{REF}} \left(\frac{R_{\mathrm{FBT}}}{R_{\mathrm{FBB}}} + 1 \right) = 0.75\,\mathrm{V} \cdot \left(\frac{33\,\mathrm{k\Omega}}{10\,\mathrm{k\Omega}} + 1 \right) \approx 3.23\,\mathrm{V} \tag{7}$$

Including the reference tolerance (0.739–0.761 V), the rail is 3.18–3.27 V, within the supply range of all ICs on the board. The divider was originally 40.2 kΩ/12 kΩ, but was changed to lower values to reduce resistor thermal noise.

Capacitors. The input and output capacitors are the 4.7 μF and 47 μF MLCCs recommended in the datasheet [MP-2]. The feed-forward capacitor is sized from the output capacitor, the module's internal 4.7 μH inductor and R_{FBT}:

$$C_{\mathrm{FF}} = \frac{\sqrt{C_{\mathrm{OUT}}\, L}}{R_{\mathrm{FBT}}} = \frac{\sqrt{47\,\mu\mathrm{F} \cdot 4.7\,\mu\mathrm{H}}}{33\,\mathrm{k\Omega}} \approx 450\,\mathrm{pF} \tag{8}$$

so the nearest standard value, 470 pF, is used.

CAN bus load. One sample (three 18-bit values) fits in a single CAN frame of at most 135 bits. At 50 Hz this is about 6.8 kbit/s, a small fraction of the bus capacity.

4.6 Interface

The board connects to the sensor interface with a 6-pin WR-WTB connector carrying V-batt, PWM, CAN-H, CAN-L and two GND pins. It takes power from the sensor interface and does not supply power to anything else.

All connectors and test points have TVS protection, as required by the shared specification [SPEC, Sec. 4.22]. V-batt uses the standard 824501261 diode, and the CAN, USB, SWD and SPI test lines use arrays from the Würth 823999 ESD kit.

A Reference material

A.1 Supporting documents and artifacts

List specifications, datasheets, drawings, repositories, schematics and test reports. Include document identifiers and revisions, or repository paths and commit hashes, so another person can recover the same source material. Use `\href{URL}{description}` for readable links.

Shared reference

[**SPEC**] Vikingur Sigurdsson, Vortex NTNU. *Team 2027 Electrical System Technical Specification Rev 23.* Document VTX-TS-002, version 0.23, 20 October 2026.

MotorController component references Manufacturer documentation consulted on 26 October 2026.

[**MC-1**] ON semiconductor. *NCP81296MNTXG eFuse for overcurrent protection and kill-switch enforcement.*, datasheet, revision 12. Datasheet and design procedure guidelines.
[**MC-2**] Texas Instruments. *TPS2640 42-V, 2-A eFuse*, datasheet, revision A. Reference for TPS26400PWPR.
[**MC-3**] Würth Elektronik. *173950336 / 3.3V PSU / MagI3C WPME-FDSM module*, Version 4.0 sep 2025
[**MC-4**] Texas Instruments. *SLVRBG7 / TPS2640 Design Calculator*, Version 01.00.00.00, Sep 2020

BMS component references Manufacturer documentation consulted on 26 October 2026.

[**BMS-1**] STMicroelectronics. *STM32C542xx microcontroller datasheet*, DS15135, revision 2, July 2026. Also: *Device errata ES0676*, revision 2, June 2026, particularly Sections 2.2.2–2.2.3.
[**BMS-2**] Texas Instruments. *BQ76942 Battery Monitor and Protector*, datasheet, revision B. Reference for the BQ7694201PFBR implementation.
[**BMS-3**] Analog Devices. *ADM3055E/ADM3057E Signal and Power Isolated CAN Transceivers*, datasheet, revision D.
[**BMS-4**] USB-C power and logic documentation: Texas Instruments, *TPS2553 current limiter* (SLVS841F), *TPS2113A power multiplexer* (SBVS045C), and *SN74LVC1G17 Schmitt-trigger buffer*; Analog Devices, *ADP3338 low-dropout regulator* (revision B).

[**BMS-5**] Texas Instruments. *TPS2640 42-V, 2-A eFuse*, datasheet, revision A. Reference for TPS26400PWPR.
[**BMS-6**] Würth Elektronik. *830108207109 / CFPX-180, 16 MHz quartz crystal*, part revision 13, 21 May 2026.
[**BMS-7**] Würth Elektronik. *830054236 / CFPX-217, 32.768 kHz watch crystal*, part revision 8, 15 April 2026.

Power Interface component references Manufacturer documentation consulted on 26 October 2026.

[**PI-1**] STMicroelectronics. *STM32C542xx microcontroller datasheet*, DS15135, revision 2, July 2026.
[**PI-2**] Texas Instruments. *TPS2640 42-V, 2-A eFuse*, datasheet, revision A. Reference for TPS26400PWPR.
[**PI-3**] Texas Instruments. *TPS25948xx 3.5V–23V, 12.2m, 8A eFuse With Bi-Directional Power Delivery*, datasheet. Reference for TPS259480LYWP.
[**PI-4**] ON semiconductor. *NCP81296MNTXG eFuse for overcurrent protection and killswitch enforcement.*, datasheet, revision 12. Datasheet and design procedure guidelines.
[**PI-5**] Analog Devices. *ADM3055E/ADM3057E Signal and Power Isolated CAN Transceivers*, datasheet, revision D.
[**PI-6**] Texas Instruments. *TCAN341x 3.3-V CAN FD Transceivers With Standby Mode and ±58 V Bus Standoff*, ,datasheet, revision A.
[**PI-7**] Texas Instruments. *TMCS1100 1% High-Precision, Basic Isolation Hall-Effect Current Sensor With ±600-V Working Voltage*, datasheet, revision C.
[**PI-8**] Würth Elektronik. *WR-CRD Micro SD Card Connector*, datasheet, revision 002.000, Sept. 2022.

Sensor Interface component references Manufacturer documentation consulted on 31 October 2026.

[**SI-1**] STMicroelectronics. *STM32H753xI microcontroller datasheet*, DS12117, revision 10, January 2026. Reference for STM32H753VIT6.
[**SI-2**] Texas Instruments. *TPS2640 42-V, 2-A eFuse*, datasheet, revision A. Reference for TPS26400PWPR, protecting the input supply to the 3.3 V and 5 V converters.
[**SI-3**] Texas Instruments. *TPS25948xx eFuse with bidirectional power delivery*, datasheet. Reference for TPS259481AYWPR, protecting the Nucleus 1000 power supply.
[**SI-4**] Würth Elektronik. *173020536 / MagI3C WPME-FDSM fixed step-down regulator module*, datasheet. Reference for the 5 V, 2 A supply.
[**SI-5**] Würth Elektronik. *173950336 / MagI3C WPME-FDSM fixed step-down regulator module*, datasheet. Reference for the 3.3 V supply.
[**SI-6**] Texas Instruments. *ISO1452 isolated full-duplex RS-485 and RS-422 transceiver*, datasheet. Reference for ISO1452BDW in the sensor communication interfaces.
[**SI-7**] Texas Instruments. *TCAN341x 3.3-V CAN FD Transceivers With Standby Mode and ±58 V Bus Standoff*, datasheet, revision A. Reference for TCAN3414DR on the vehicle and dedicated magnetometer CAN buses.
[**SI-8**] Würth Elektronik. *18012x15411x / WPME-CDIS two-channel digital isolator*, datasheet. Reference for 18012115411L.
[**SI-9**] Microchip Technology. *LAN8742A/LAN8742Ai Small Footprint RMII 10/100 Ethernet Transceiver*, DS00001989A. Reference for LAN8742AI-CZ-TR.
[**SI-10**] Würth Elektronik. *74990111217 RJ45 connector with integrated Ethernet magnetics*, datasheet. Reference for the Ethernet cable interface.
[**SI-11**] Würth Elektronik. *830055299 / CFPX-180, 25 MHz quartz crystal*, part revision 13, 21

May 2026. Reference for the Ethernet PHY crystal.
[SI-12] Safran Sensing Technologies Norway (Sensonor). *STIM300 Inertial Measurement Unit*, manufacturer datasheet. Reference for the STIM300 sensor interface.
[SI-13] Nortek. *Nucleus 1000 underwater navigation sensor package*, product documentation and links to the operations and integration manual. Reference for the Nucleus 1000 sensor interface.
[SI-14] Blue Robotics. *Bar High-Resolution Depth/Pressure Sensors Guide*, manufacturer integration guide. Reference for the Bar30 pressure sensor, electrical connections and I2C interface.
[SI-15] Würth Elektronik. *629722000214 / WR-COM USB 2.0 Type-C Horizontal Receptacle SMT*, datasheet, revision 002.000, 4 September 2025. Reference for the USB-C connector.

Orin BOB component references Manufacturer documentation consulted on 31 October 2026.

[OB-1] Würth Elektronik. *18024x15401x WPME-CDIP 4-channel digital isolator with integrated 0.65 W isolated DC/DC*, datasheet, version 1.1, November 2023. Reference for 18024115401H and 18024215401H.
[OB-2] Würth Elektronik. *18032x11 WPME-CDI2C 2-channel I2C isolator*, datasheet, version 1.0, April 2026. Reference for 18032011.
[OB-3] Analog Devices. *ADM3055E/ADM3057E Signal and Power Isolated CAN Transceivers*, datasheet, revision D.
[OB-4] NVIDIA. *Jetson Linux Developer Guide: Controller Area Network (CAN)*, release r36.4.3. CAN pin-out of the Jetson AGX Orin 40-pin header.
[OB-5] JetsonHacks. *NVIDIA Jetson AGX Orin J30 GPIO expansion header pinout*. Based on the NVIDIA Jetson AGX Orin DevKit Carrier Board Specification.
[OB-6] NXP Semiconductors. *UM10204 I2C-bus specification and user manual*, revision 7.0. Pull-up resistor and rise-time equations.

Orin Interface component references Manufacturer documentation consulted on 31 October 2026.

[OI-1] Texas Instruments. *TPS25948xx 3.5V–23V, 12.2mΩ, 8A eFuse With Bi-Directional Power Delivery*, datasheet, revision D. Reference for TPS259482AYWPR.
[OI-2] Analog Devices. *LT8357 60V 2MHz Low-IQ Boost, SEPIC and Flyback Controller*, datasheet, revision A. Reference for LT8357EMSE-PBF.
[OI-3] Texas Instruments. *TPS23861 IEEE 802.3at Quad Port Power-over-Ethernet PSE Controller*, datasheet, revision I. Reference for TPS23861PWR.
[OI-4] Texas Instruments E2E forum. *TPS23861: overcurrent setting*. Auto-mode I2C pin handling.

Magnetometer PCB component references Manufacturer documentation consulted on 31 October 2026.

[MP-1] STMicroelectronics. *STM32C542xx microcontroller datasheet*, DS15135, revision 2, July 2026.
[MP-2] Würth Elektronik. *171930601 MagI3C WPME-VDMM variable step-down micromodule*, datasheet, version 2.0, July 2023. Reference for the V-batt to 3.3 V supply.
[MP-3] Texas Instruments. *TCAN341x 3.3-V CAN FD Transceivers With Standby Mode and ±58 V Bus Standoff*, datasheet, revision A.

[**MP-4**] Würth Elektronik. *830207659609, 48 MHz oscillator*, datasheet. Reference for the HSE clock.

[**MP-5**] Würth Elektronik. *830205028309, 32.768 kHz oscillator*, datasheet. Reference for the LSE clock.

[**MP-6**] MEMSIC. *MMC5983MA ±8 Gauss, High Performance 3-axis Magnetic Sensor*, datasheet, revision A. Reference for the magnetometer, SPI interface, SET/RESET function and noise versus bandwidth.

A.2 Terms and abbreviations

Define project-specific terms, symbols, and abbreviations used in the document.

A.3 Supplementary details

Place lengthy calculations, additional figures, or configuration examples here when they would interrupt the main narrative. Remove unused subsections.